Concept 4 | Advanced Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Differences

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping

odd/even index $\longrightarrow$ separate rules

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Classified Practice **Advanced Sequences**

Q001 **Written**

Find the general term of the sequence $\{2, 5, 12, 23, 38, 57, \dots\}$.

Classified Practice **Advanced Sequences**

Q002 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, then a_n is:

Choose the correct option.

A $n^2 - 3$
B $(n+1)^2$
C $a_n = n^2$
D $n^2 + 3$

Q003 **MCQ**

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, then a_n is:

Choose the correct option.

A $\frac{3n^2-n+4}{2}$

B $a_n = \frac{3n^2-7n+8}{2}$

C $\frac{3n^2-7n+16}{2}$

D $\frac{3n^2-7n+6}{2}$

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Q004 MCQ

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, then a_n is:

Choose the correct option.

A $a_n = 3n^2 - 8n + 12$
B $3n^2 - 8n + 13$
C $3n^2 - 8n + 9$
D $3n^2 - 2n + 7$

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Q005 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, find a_5:

Choose the correct option.

A -25

B 25

C 30

D 23

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Q006 MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, find a_6:

Choose the correct option.

A 37

B 38

C 34

D -37

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Q007 *MCQ*

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, find a_6:

Choose the correct option.

A 73
B 70
C -72
D 72

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Q008 Written

Find the general term of the sequence $\{2, 10, 24, 44, 70, 102, \ldots\}$.

Classified Practice **Advanced Sequences**

Q009 **Written**

Find the general term of the sequence $\{1, 6, 15, 28, 45, 66, \ldots\}$.

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Q010 MCQ

Select the correct option: If $a_1 = 3$ and $b_k = 5k + 4$, then a_n is:

Choose the correct option.

A $a_n = \frac{(n+1)(5n-2)}{2}$

B $\frac{n(5n+3)}{2}$

C $\frac{5n^2+3n-6}{2}$

D $\frac{(n+2)(5n+3)}{2}$

Q011 MCQ

Select the correct option: If $a_1 = 4$ and $b_k = 7k - 3$, then a_n is:

Choose the correct option.

A $\frac{7n^2-13n+18}{2}$

B $\frac{7n^2-13n+8}{2}$

C $\frac{7n^2+n+8}{2}$

D $a_n = \frac{7n^2-13n+14}{2}$

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Q012 MCQ

Select the correct option: If $a_1 = 6$ and $b_k = 2(2k+3)$, then a_n is:

Choose the correct option.

A $2(n+1)(n+3)$
B $a_n = 2n(n+2)$
C $2\left(n^2+2n+2\right)$
D $2\left(n^2+2n-1\right)$

Classified Practice **Advanced Sequences**

Q013 **Written**

Find the general term of the sequence $\{5, 8, 9, 8, 5, 0, \dots\}$.

Classified Practice **Advanced Sequences**

Q014 **Written**

Find the general term of the sequence $\{2, 3, 6, 15, 42, 123, \ldots\}$.

Classified Practice **Advanced Sequences**

Q015 **Written**

Find the general term of the sequence $\{1, 2, 5, 14, 41, 122, \ldots\}$.

Classified Practice **Advanced Sequences**

Q016 **Written**

Find the general term of the sequence $\{5, 7, 11, 19, 35, \ldots\}$.

Classified Practice **Advanced Sequences**

Q017 **Written**

Find the general term of the sequence $\{2, 3, 1, 5, -3, 13, \ldots\}$.

Classified Practice **Advanced Sequences**

Q018 **Written**

Find the general term of the sequence $\{5, 4, 7, -2, 25, -56, \dots\}$**.**

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Q019 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = k^2$, then a_n is:

Choose the correct option.

A $a_n = \frac{(n+1)(2n^2-5n+6)}{6}$

B $\frac{2n^3-3n^2+n+12}{6}$

C $\frac{2n^3-3n^2+n-12}{6}$

D $\frac{(n+2)(2n^2-n+3)}{6}$

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Q020 Challenge

An oncology team records tissue volumes $1, 3, 7, 13, 21, 31, \ldots$ in mm^3. Find the general formula a_n and calculate a_{12} to decide whether the 130 mm^3 threshold is exceeded.

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Q021 **Written**

For groups with $a_j = 2 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j, find the first term of group 10.

Q022 **Written**

For groups with $a_j = 1 + (j-1)2$ and $c_j = 2n - 1$ terms in group j, find the first term of group 10.

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Q023 **Written**

For groups with $a_j = 1 + (j - 1)1$ and $c_j = n$ terms in group j, find the first term of group 10.

Q024 MCQ

Select the correct option: find the first term of group 3:

Choose the correct option.

A 6

B 4

C 5

D 3

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Q025 MCQ

Select the correct option: find the first term of group 6:

Choose the correct option.

A 77
B 108
C 78
D 79

Q026 MCQ

Select the correct option: find the first term of group 9:

Choose the correct option.

A 38

B 36

C 45

D 37

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Q027 MCQ

Select the correct option: find the first term of group 4:

Choose the correct option.

A 48

B 30

C 31

D 29

Q028 MCQ

Select the correct option: find the first term of group 7:

Choose the correct option.

A 21
B 28
C 22
D 23

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Q029 MCQ

Select the correct option: find the first term of group 10:

Choose the correct option.

A 247

B 245

C 300

D 246

Q030 MCQ

Select the correct option: find the first term of group 4:

Choose the correct option.

A 20

B 21

C 19

D 32

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Q031 MCQ

Select the correct option: find the first term of group 7:

Choose the correct option.

A 138
B 110
C 111
D 109

Q032 MCQ

Select the correct option: find the first term of group 10:

Choose the correct option.

A 163

B 200

C 164

D 165

Classified Practice **Advanced Sequences**

Q033 **MCQ**

Select the correct option: find the first term of group 5:

Choose the correct option.

A 58

B 59

C 57

D 78

Q034 MCQ

Select the correct option: find the first term of group 8:

Choose the correct option.

A 128

B 100

C 101

D 99

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Q035 **Written**

For groups with $a_j = 1 + (j-1)3$ and $c_j = 3n - 1$ terms in group j, find the first term of group 10.

Q036 MCQ

Select the correct option: find the first term of group 5:

Choose the correct option.

A 30

B 28

C 39

D 29

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Q037 MCQ

Select the correct option: find the first term of group 8:

Choose the correct option.

A 89
B 90
C 88
D 110

Q038 **Written**

For groups with $a_j = 2 + (j-1)2$ and $c_j = 2n - 1$ terms in group j, derive the sum of the terms in group n.

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Q039 **Written**

For the same grouped sequence, find the sum of the terms in group 10.

Q040 **Written**

For the same grouped sequence, find the sum of the terms in group 10**.**

Classified Practice **Advanced Sequences**

Q041 Written

For the same grouped sequence, find the sum of the terms in group 10.

Q042 **Written**

In the grouped arithmetic sequence, locate the value 88: state its group and its position within that group.

Classified Practice **Advanced Sequences**

Q043 **Written**

In the grouped arithmetic sequence, locate the value 101: state its group and its position within that group.

Q044 **Written**

In the grouped arithmetic sequence, locate the value 124: state its group and its position within that group.

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Q045 Written

In the grouped arithmetic sequence, locate the value 100: state its group and its position within that group.

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Q046 Challenge

For the grouped sequence $2, 6 \mid 10, 14, 18 \mid 22, 26, 30, 34 \mid \ldots$, solve: (a) the first term of Group 12; (b) the sum in Group n; (c) the group and position of 202; (d) the total through Group 12.

Classified Practice **Advanced Sequences**

Quiz MCQ 1 Concept Quiz · MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k + 1$, then a_n is:

Choose the correct option.

A $a_n = \frac{3n^2-n+2}{2}$

B $\frac{3n^2-n+4}{2}$

C $\frac{n(3n-1)}{2}$

D $\frac{3n^2+5n+4}{2}$

Quiz MCQ 2 **Concept Quiz · MCQ**

Select the correct option: If $a_1 = 3$ and $b_k = (-2)^{k-1}$, then a_n is:

Choose the correct option.

A $\frac{(-2)^n+2}{6}$

B $-\frac{(-2)^n-10}{3}$

C $a_n = \frac{(-2)^n+20}{6}$

D $\frac{(-2)^n+38}{6}$

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Quiz MCQ 3 Concept Quiz · MCQ

Select the correct option: find the first term of group 9:

Choose the correct option.

A 195
B 196
C 194
D 243

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: the value 29 is in:

Choose the correct option.

A Group 5, position 2

B Group 5, position 1

C Group 6, position 1

D Group 4, position 2

Classified Practice **Advanced Sequences**

Quiz WRITTEN 5 Concept Quiz · Written

A sequence has $a_1 = 1$ and difference sequence $b_k = 4k - 1$. Find the required general term a_n.

Quiz WRITTEN 6 **Concept Quiz · Written**

A sequence has $a_1 = 2$ and difference sequence $b_k = (-2)^{k-1}$. Find the required general term a_n.

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Quiz WRITTEN 7 Concept Quiz · Written

For groups with $a_j = 5 + (j - 1)3$ and $c_j = n$ terms in group j, find the first term of group 12.

Quiz WRITTEN 8 **Concept Quiz · Written**

In the grouped arithmetic sequence, locate the value 66: state its group and its position within that group.

Answer key

Use the question number shown on each page.

Q001 $a_n = 2n^2 - 3n + 3$

Q002 C

Q003 B

Q004 A

Q005 B

Q006 A

Q007 D

Q008 $a_n = n(3n-1)$

Q009 $a_n = n(2n-1)$

Q010 A

Q011 D

Q012 B

Q013 $a_n = -n(n-6)$

Q014 $a_n = \dfrac{3^n + 9}{6}$

Q015 $a_n = \dfrac{3^n + 3}{6}$

Q016 $a_n = 2^n + 3$

Q017 $a_n = \dfrac{(-2)^n + 14}{6}$

Q018 $a_n = -\dfrac{(-3)^n - 57}{12}$

Q019 A

Q020 $a_n = n^2 - n + 1, \quad a_{12} = 133;$ surgery required

Q021 164

Q022 163

Q023 46

Q024 B

Q025 C

Q026 D

Q027 B

Q028 C

Q029 D

Q030 A

Q031 B

Q032 C

Q033 A

Q034 B

Q035 379

Q036 D

Q037 A

Q038 $2(2n-1)\left(n^2 - n + 1\right)$

Q039 3439

Q040 505

Q041 12209

Q042 Group 7, position 8

Q043 Group 8, position 2

Q044 Group 16, position 4

Q045 Group 5, position 8

Q046 (a) 310; (b) $2(n+1)\left(n^2+2n-1\right)$; (c) Group 9, position 7; (d) 16200

Quiz MCQ 1 A

Quiz MCQ 2 C

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $a_n = 2n^2 - 3n + 2$

Quiz WRITTEN 6 $a_n = \dfrac{(-2)^n + 14}{6}$

Quiz WRITTEN 7 203

Quiz WRITTEN 8 Group 5, position 3